

Playbook: Onboarding a New Language into ABAP Assist

Target Landscape: SAP S/4HANA / ABAP Platform 7.5+

Audience: ABAP Developers, Translation Specialists, SAP Release Engineers

Status: Standard Operating Procedure (SOP)

1. Executive Summary & Architecture Overview

Thanks to the decoupled multi-lingual architecture implemented in ABAP Assist:

- **Zero Core Code Modifications:** The core AI engine (`ZCLCA_ABAP_ASSIST_AIUTIL`) dynamically resolves the user's logon language (`sy-langu`) from SAP standard table `T002T` and automatically instructs Google Cloud Vertex AI / Gemini via `{llm_convo_lang}`.
- **Extensible Text Table Architecture:** Prompt templates are stored in a dedicated text table `ZCA_PROMPT_TMPLT`, ready to accept any standard SAP language key (`SPRAS`).
- **Built-in Fallback Safeguard:** If a translation is missing for any specific string, the application automatically falls back to English (`'E'`) without throwing runtime errors.

Onboarding any new language (e.g., **German DE**, **French FR**, **Spanish ES**, **Chinese ZH**, **Portuguese PT**, etc.) requires **only localization configuration, text translation, and data maintenance**—with **zero changes to core ABAP classes**.

2. Step-by-Step Implementation Guide

Step 1: System & Language Prerequisites Verification

Before starting translation, verify that the target language is installed and active in the SAP system:

1. **Verify Language in SMLT (Language Management):**
 - Open transaction `SMLT`.
 - Confirm that the target language (e.g., German `DE / D`, French `FR / F`, Spanish `ES / S`) is installed and active.

2. Verify Table **T002T** Language Name:

- Open transaction **SE16N** or query **T002T**:

SQL

```
SELECT sptxt FROM t002t
WHERE sprsl = '<TARGET_LANG_KEY>' -- e.g. 'D' for German
AND spras = 'E' -- English text descriptor
```

- Confirm that **sptxt** returns the valid English name (e.g., 'German', 'French', 'Spanish'). This value is dynamically passed to Gemini as **{llm_convo_lang}**.
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Step 2: Create Transport Requests

In transaction **SE01** / **SE09** / **SE10**, create the following transport requests:

1. Language Translation Transport (TR 1):

- Type: **Transport of Copies** or **Workbench Request**
- Description: **LANG <language_code> Translation for ABAP Assist** (e.g. LANG deDE German Translation)

2. Customizing / Table Content Transport (TR 2):

- Type: **Workbench / Customizing Request**
 - Description: **Data: Localized Prompt Templates for ZCA_PROMPT_TMPLT (<Language>)**
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Step 3: Translate Repository Artifacts in Transaction **SE63**

Translate all ABAP Assist artifacts from **English (enUS)** to the **Target Language**:

A. Message Class **ZCA_ABAP_ASSIST** (Messages **001 – 014**)

- **Transaction:** **SE63** -> *Short Texts* -> *ABAP Texts* -> *T100 Messages* (or enter **ZCA_ABAP_ASSIST**).
- **Source Language:** **enUS** / **English**
- **Target Language:** Target Language (e.g., **deDE** / **German**)
- **Messages to Translate:**

Message #	English Text (E)	Target Translation Guidelines
001	Object/namespace is not whitelisted for ABAP Assistant	Translate to local language
002	ABAP Assist currently not supporting in Webgui	Translate to local language
003	Missing Genie Auth for the feature &1.	Keep &1 placeholder intact
004	Only single Org entry supported for access control in current version.	Translate to local language
005	Feature is either disabled or not configured in table ZCAC_GENIE_ACCES.	Keep table name ZCAC_GENIE_ACCES intact
006	Maintain Model Key in the team &1	Keep &1 placeholder intact
007	Package is not whitelisted. Approval required before adding std package.	Translate to local language
008	AI ABAP Assistant Context Setup Successful!	Translate to local language
009	AI ABAP Assist Context Setup failed. Please contact Google ABAP SDK Team!	Translate to local language
010	AI ABAP Assistant Setup Deleted!	Translate to local language

Message #	English Text (E)	Target Translation Guidelines
011	AI ABAP Assis Setup Deletion failed. Please contact Google ABAP SDK Team!	Translate to local language
012	Could not determine system client category.	Translate to local language
013	This tool should not be used in Production or SAP Reference systems.	Translate to local language
014	Invalid regular expression	Translate to local language

B. GUI Presentation Class Text Pool: ZCLCA_ABAP_ASSIST_UI

- **Transaction:** SE63 -> Short Texts -> ABAP Texts -> CPUB / Text Pools -> Enter ZCLCA_ABAP_ASSIST_UI=====CP.
- **Translate Text Symbols (001 through 019):**

Symbol	English Text (E)	Purpose in UI
text-001	(Reserved / Title)	UI Container Title
text-003	Help me suggest.	Suggest code prompt fallback
text-004	Help me explain.	Explain code prompt fallback
text-005	Help me review / test.	Code review & unit test prompt fallback
text-006	Default prompt	Prompt builder placeholder
text-007	No response generated.	Response handler error message

Symbol	English Text (E)	Purpose in UI
text-008	New Chat	New Chat button (+ { text-008 })
text-009	Recent Chats	Navigation sidebar heading
text-010	Response formatting	HTML parsing and response formatting
text-011	Accept	Code block Accept button
text-012	Copy	Code block Copy to clipboard button
text-013	Welcome to ABAP Assist	Main welcome banner heading
text-014	Select a quick action	Startup tiles subtitle
text-015	Explain	Quick Action button: Explain Code
text-016	Review	Quick Action button: Code Review
text-017	Suggest	Quick Action button: Suggest Improvement
text-018	Unit Test	Quick Action button: ABAP Unit Test
text-019	Select Model	Model selection dropdown label

C. Report Title & Additional Class Text Pools

- **ZR_AI_ABAP_ASSIST_CONTEXT_MENU** (Program Title & Selection Texts):
 - Translate title: Gemini Powered SAP ABAP Editor -> Localized Report Title (e.g. Gemini-basierter SAP ABAP-Editor).
- **ZCLCA_ABAP_ASSIST_ADT_RESOURCE** (ADT REST Endpoint Metadata):
 - text-001: Localized metadata descriptor.
- **ZCLCA_ABAP_ASSIST_ADT_RES_CONT** (ADT Content REST Endpoint):
 - text-001: Localized content metadata descriptor.

- **ZCLCA_AI_ASSISTANT_ENHANCEMENT** (Context Menu Integration):
 - **text-009**: Localized fallback block descriptor.
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D. Message Long Text Documentation (DOCU NA)

- **Transaction**: **SE63** -> *Long Texts* -> *Documentation* -> **NA** -> **NAZCA_ABAP_ASSIST014**.
 - Translate the diagnosis, system response, and procedure for Message 014.
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Step 4: Maintain Localized Prompt Templates in **ZCA_PROMPT_TMPLT**

Prompt templates for the new language must be maintained in text table **ZCA_PROMPT_TMPLT**.

Method A: Via Table Maintenance (**SM30**)

1. Open transaction **SM30**.
2. Enter table name: **ZCA_PROMPT_TEMPL** and click **Maintain**.
3. Select each template row (**T1**, **T2**, **T3**, **T4**) -> navigate to menu **Edit -> Translation**.
4. In the translation popup, choose the target language and enter the localized **Description** and **Template Text**.
5. Save changes.

Method B: Via ABAP Data Migration Script

Alternatively, insert the records programmatically in SE38:

None

```
DATA: lt_new_lang_templates TYPE STANDARD TABLE OF zca_prompt_tmplt.
```

```
lt_new_lang_templates = VALUE #(
  ( mandt      = sy-mandt
    spras      = '<TARGET_LANG_KEY>' " e.g., 'D' for German, 'F' for French
    id         = 'T1'
    description = '<Localized Description 1>'
    template   = '<Localized Prompt Template with {use-case}>' )
  ( mandt      = sy-mandt
    spras      = '<TARGET_LANG_KEY>'
    id         = 'T2'
    description = '<Localized Description 2>'
    template   = '<Localized Prompt Template with {code}>' )
  ( mandt      = sy-mandt
    spras      = '<TARGET_LANG_KEY>' )
```

```

id          = 'T3'
description = '<Localized Description 3>'
template    = '<Localized Prompt Template with {code}>' )
( mandt     = sy-mandt
  spras     = '<TARGET_LANG_KEY>'
  id        = 'T4'
  description = '<Localized Description 4>'
  template   = '<Localized Prompt Template with {code}>' )
).
```

```

MODIFY zca_prompt_tmplt FROM TABLE lt_new_lang_templates.
COMMIT WORK.
```

 **Prompt Variable Tokens:** Ensure template placeholders (`{use-case}`, `{code}`) are preserved verbatim in the localized template text so that string interpolation continues working seamlessly.

Step 5: Capture Objects in Transport Requests

1. Capture Translation Objects (in Translation TR):

- Using transaction **SE03** -> *Translation Organizer* -> *Collect Translations* for package **ZCA_ABAP_ASSIST** and target language.
- Verify the object list contains:
 - **LANG MESS ZCA_ABAP_ASSIST...**
 - **LANG REPS ZR_AI_ABAP_ASSIST_CONTEXT_MENU**
 - **LANG REPT ZR_AI_ABAP_ASSIST_CONTEXT_MENU**
 - **LANG REPT ZCLCA_ABAP_ASSIST_UI=====CP**
 - **LANG REPT ZCLCA_ABAP_ASSIST_ADT_RESOURCECP**
 - **LANG REPT ZCLCA_ABAP_ASSIST_ADT_RES_CONTCP**
 - **LANG REPT ZCLCA_AI_ASSISTANT_ENHANCEMENTCP**

2. Capture Prompt Table Entries (in Workbench / Customizing TR):

- In **SE01**, add the table entries object:
 - **PGMID:** **R3TR**
 - **Object:** **TABU**
 - **Object Name:** **ZCA_PROMPT_TMPLT**
 - **Table Keys** (): Enter ***** (or specific language key e.g. **001D***).
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Step 6: Prompt Maintenance using Prompt Manager

Update below Prompt Ids using Prompt Manager(TCode: ZCA_PROMPT_MGR)

- EXPLANATION
- CODE_REVIEW
- CODE_SUGGEST
- UNIT_TEST
- CODE_GEN

Please refer to the section **Update of Prompts using tcode : ZCA_PROMPT_MGR** in **Product Documentation** to update/modify existing prompts with new text. Always refer to the **AI ABAP Assistant Prompts sheet** for the most recent prompts.

Step 7: Quality Assurance & Verification Test Matrix

Log into SAP in the **Target Language** and execute the standard 6-step test suite:

Test Case ID	Test Scenario	Execution Steps	Expected Result
TC-01	UI Localization	Log in with the target language. Launch ABAP Assist in SE38 / Debugger.	Banner title, "+ New Chat", Recent Chats, and quick action buttons ("Explain", "Review", "Suggest", "Unit Test") render in the target language.
TC-02	Prompt Templates	Inspect startup tiles and template dropdown in UI / ADT.	Localized descriptions and templates from ZCA_PROMPT_TMPLT appear correctly.
TC-03	Conversational LLM	Send an ABAP coding query in the target language (e.g., request ALV code or explanation).	Gemini responds with explanations, analysis, and ABAP code comments entirely in the target language.
TC-04	System Messages	Trigger authorization or validation errors	Message class ZCA_ABAP_ASSIST

Test Case ID	Test Scenario	Execution Steps	Expected Result
		(e.g. non-whitelisted package).	(001–014) outputs localized text.
TC-05	ADT Plugin Integration	Connect to ADT under the target language locale. Launch ABAP Assist view.	ADT plugin receives localized prompt templates and metadata.
TC-06	Fallback Verification	Temporarily delete or omit one prompt template for the language.	The application gracefully falls back to the English baseline (' E ') without crashing.

Step 8: Release & Deployment Sequence

1. Release the **Customizing / Table Content Transport (TABU)**.
 2. Release the **Language Translation Transport (LANG)**.
 3. In downstream systems (SIT -> UAT -> PROD), import transports in the standard order.
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3. Onboarding Summary & Effort Estimation

Phase	Activity	Responsible Role	Estimated Effort
Phase 1	System Language Check (SMLT / T002T)	BASIS / Developer	~ 0.5 hours
Phase 2	SE63 Translation (Messages, Text Pools, Titles)	Functional / Translator	~ 2 - 4 hours
Phase 3	Prompt Template Maintenance (ZCA_PROMPT_TMPLT)	Developer / AI Lead	~ 1 - 2 hours

Phase	Activity	Responsible Role	Estimated Effort
Phase 4	Transport Packaging & Table Key Capture (TABU)	ABAP Developer	~ 0.5 hours
Phase 5	QA Testing & Validation (6-Point Matrix)	QA / Developer	~ 2 - 3 hours
Total	End-to-End Language Onboarding		**~ 1 Developer Day**